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[REDACTED] make a preliminary ecological appraisal of a potential wind farm site at Riverdale, Raharney, County Westmeath.

2 METHODOLOGY

Preliminary walkover surveys were conducted on the 24th and 26th of March 2021 to determine the characteristics of the potential wind farm site. Habitats on site were identified according to *Fossitt (2000) Guide to Habitats in Ireland* and provisionally assessed with regards their vegetation, condition, and conservation status. A multidisciplinary ecological walkover survey was undertaken on the 21st of April 2022 with a particular focus on terrestrial mammals. Comprehensive bird surveys commenced in Spring 2021 and are ongoing at the site. The results of these surveys are not presented in this report.

The Office of the Planning Regulator (OPR) source-pathway-receptor (S-P-R) model¹ was used to determine the designated sites within the potential zone of influence (ZOI) of the potential wind energy site. Considering the scale of the potential wind farm development, the ZOI is considered to potentially affect designated sites within a 5km radius of the site. Furthermore, an assessment of designated sites and sensitive watercourses occurring downstream of the site was carried out to determine connectivity between the survey site and sites designated for nature conservation. The National Roads Authority Guidelines for Assessment of Ecological Impacts of National Roads Schemes² was used to determine the ecological value of the habitats within the potential wind farm site.

¹ Office of the Planning Regulator (2021) Appropriate Assessment Screening for Development Management. OPR Practice Note PN01

² Guidelines for Assessment of Ecological Impacts of National Roads Schemes. National Roads Authority, June 2009.

3 RESULTS

3.1 DESIGNATED SITES

Sites designated for Nature Conservation within the ZoI of the potential wind energy site are presented in Table 1 below and are illustrated in Figure 1. Those sites identified within a 5km radius of the potential wind farm site include the River Boyne and River Blackwater SAC (NPWS Site Code: 002299), Mount Hevey Bog SAC / pNHA (NPWS Site Code: 002342), River Boyne and River Blackwater SPA (NPWS Site Code: 004232), and the Royal Canal pNHA (NPWS Site code: 002103).

 occurs within the central part of the site and provides surface water to the River Boyne and River Blackwater SAC circa 1.9km downstream. The Boyne and its tributaries provide excellent breeding conditions for fish species, including salmon (*Salmo salar*) and brown trout (*Salmo trutta*) (see Table 1). The River Boyne and River Blackwater SPA also occurs circa 1.9km downstream via the Craddanstown Stream. This SPA is designated for the protection of kingfisher (*Alcedo atthis*), a Species listed on Annex I of the EU Birds Directive.

Mount Hevey Bog SAC / pNHA is situated approximately 3.8km to the south of the proposed wind farm site. It is hydrologically isolated from the site. The SAC is of international importance due to the occurrence of high quality raised bog habitats.

The Royal Canal pNHA is situated 3.6km south of the development site. The canal has been proposed as a natural heritage area (pNHA). The canal is situated upstream from the River Boyne and hosts a variety of bird and other species inside and along its banks.

Table 1: Designated sites within the zone of influence of the potential wind farm.

Site Name	Site Code	Location / Connectivity
River Boyne and River Blackwater SAC	002299	The SAC occurs circa 1.7km to the south and is hydrologically connected circa 1.9km downstream via the Craddanstown Stream.
River Boyne and River Blackwater SPA	004232	The SPA occurs circa 1.7km to the south and is hydrologically connected circa 1.9km downstream via the Craddanstown Stream.
Mount Hevey Bog SAC / pNHA	002342	The SAC occurs circa 3.8km to the south and is hydrologically isolated from the potential wind farm site.
Royal Canal pNHA	002103	The pNHA occurs circa 3.6km to the south and is hydrologically isolated from the potential wind farm site.

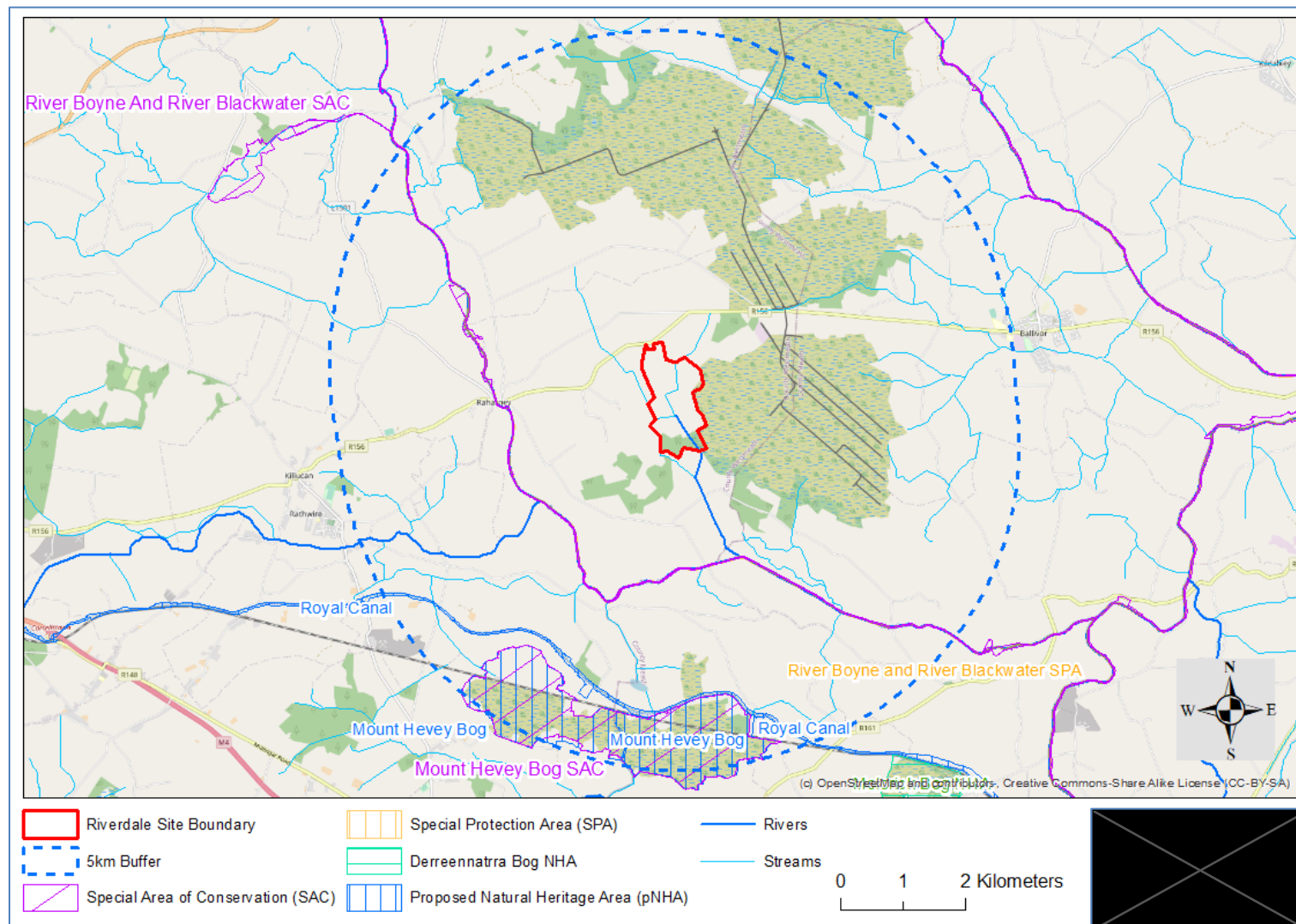


Figure 1: Map showing the location of designated sites within 5km of the potential wind farm site and those occurring downstream (Note that the River Boyne and River Blackwater SAC and SPA tentirely overlap along the course of the river).



3.2 HABITATS

A map showing the general distribution of the main habitats within the site is presented in Figure 2. The following habitats were recorded during the site visit undertaken during April 2022.

- **Improved agricultural grassland (GA1):** This habitat is intensively managed for agriculture and is the dominant habitat within the site. Improved grassland occurs mainly toward the northern and mid-western parts of the site. Perennial ryegrass (*Lolium perenne*) is the dominant species and white clover (*Trifolium repens*) occurs in abundance. The majority of the improved pastures are divided by deep drainage channels (FW4) circa 2.5m deep and hedgerows (WL1) or treelines (WL2). The habitat is relatively species poor being subject to nutrient inputs and intensive grazing.
Evaluation: This habitat is of low ecological importance.
- **Wet grassland (GS4):** A small area of wet grassland occurs at the southern part of the survey area. There is a range of rank grasses occurring within this habitat, with a dominance of rushes (*Juncus effusus*) and bramble (*Rubus fruticosus* agg.). Gorse (*Ulex europaeus*), birch (*Betula pubescens*), and willow (*Salix* sp.) have been cleared from this area.
Evaluation: This habitat is of local importance, lower value.
- **Tilled land (BC3):** A large plot has been recently tilled at the north-eastern part of the site. Tilled land is of low ecological value, whilst the production of arable crops represents intensively managed agricultural land.
Evaluation: This habitat is of low ecological importance.
- **Conifer forestry (WD4):** Two dense mature conifer plantations, consisting predominantly of Norway spruce (*Picea abies*) with Japanese larch (*Larix kaempferi*) were recorded in the central and southern parts of the survey area. Broadleaf trees were recorded around the perimeters of both plantations. These borders included an abundance of sycamore (*Acer pseudoplatanus*), birch (*Betula pubescens*), alder (*Alnus glutinosa*), and mountain ash / Rowan (*Sorbus aucuparia*). The plantation is very dense which has caused the forest floor to be void of any major biological diversity, typical of conifer plantations.. A small pocket of birch woodland occurs within the southern conifer plantation on an area of cutover bog.
Evaluation: This habitat is of low ecological importance.
- **Dense Bracken (HD1):** Dense bracken occurs circa 15m to the east of cutover bog habitat. Stacks of branches from harvested timber and some spoil (see below) occurs throughout the areas of dense bracken. This area is flat, and borders improved agricultural grassland, and semi-natural woodland.
Evaluation: This habitat is of low ecological importance.

- **Spoil and bare ground (ED2):** Some spoil heaps occur within dense bracken habitat. Much of the spoil occurring throughout the dense bracken has become overgrown with plants such as herb robert (*Geranium robertianum*), nettles (*Urtica dioica*), and bramble. No other spoil was noted during the survey.
Evaluation: This habitat is of low ecological importance.
- **(Mixed) Broadleaved woodland (WD1):** A plantation of sycamore trees, circa 4 - 5m in height is situated amongst the southernmost conifer plantation in the site. The understorey consists of broadleaved herbs and rank grasses. Other broadleaved areas include a linear buffer of mature birch between the cutover bog (see below) and dense bracken.
Evaluation: This habitat is of low ecological importance.
- **Semi-natural woodland (WN1):** Semi-natural woodland occurs in three areas at the southern part of the site as mapped in Figure 2. Birch is dominant, with hawthorn (*Crataegus monogyna*), holly (*Ilex aquilinum*), and willow (*Salix* sp.) occurring occasionally. Bramble is abundant throughout this habitat.
Evaluation: This habitat is of local importance, lower value.
- **Treelines (WL2):** Treeline habitats generally consisted of ash (*Fraxinus excelsior*) with bramble and ivy (*Hedera helix*). Treelines form the west and northeast boundaries of the Riverdale site, in addition to forming internal field boundaries. The treeline habitats at this site are considered to be locally important, providing an important refuge for a variety of species. These treelines create important wildlife corridors and form landscape connectivity to neighbouring habitats. Treeline habitats in the central region (west of the conifer plantation habitat) of the site provide suitable foraging habitat for bats.
Evaluation: This habitat is of local importance, higher value.
- **Hedgerows (WL1):** Hedgerows within the site are dominated by hawthorn and bramble. Hedgerows form part of internal and external field boundaries within the site. Hedgerows and treelines often occur together, creating suitable foraging and nesting habitats for bird species and pathways for badgers and other mammals to traverse the landscape and forage for food.
Evaluation: This habitat is of local importance, higher value.
- **Cutover bog (PB4):** The area of cutover bog area in the south-western part of site is dominated by ling heather (*Calluna vulgaris*) with some stunted lodgepole pine (*Pinus contorta*) scattered throughout. This area provides good opportunities for common nesting birds.
Evaluation: This habitat is of local importance, higher value.
- **Lowland / depositing rivers (FW2):** There are two natural watercourses within the footprint of the proposed development that correspond with this habitat type, the Grange More stream and

Craddanstown Stream. Grange More Stream flows southerly from the upper part of the central conifer plantation of the site and continues to the Craddanstown Stream. The Craddanstown Stream then converges with the Deel river approximately 2km downstream from the potential wind farm site. This waterbody then flows through the most northern plantation. Both watercourses were circa 3m deep and 3m wide. Most field drains within the site were relatively dry.

Evaluation: This habitat is of international importance due to its direct connectivity to the River Boyle SAC / SPA.

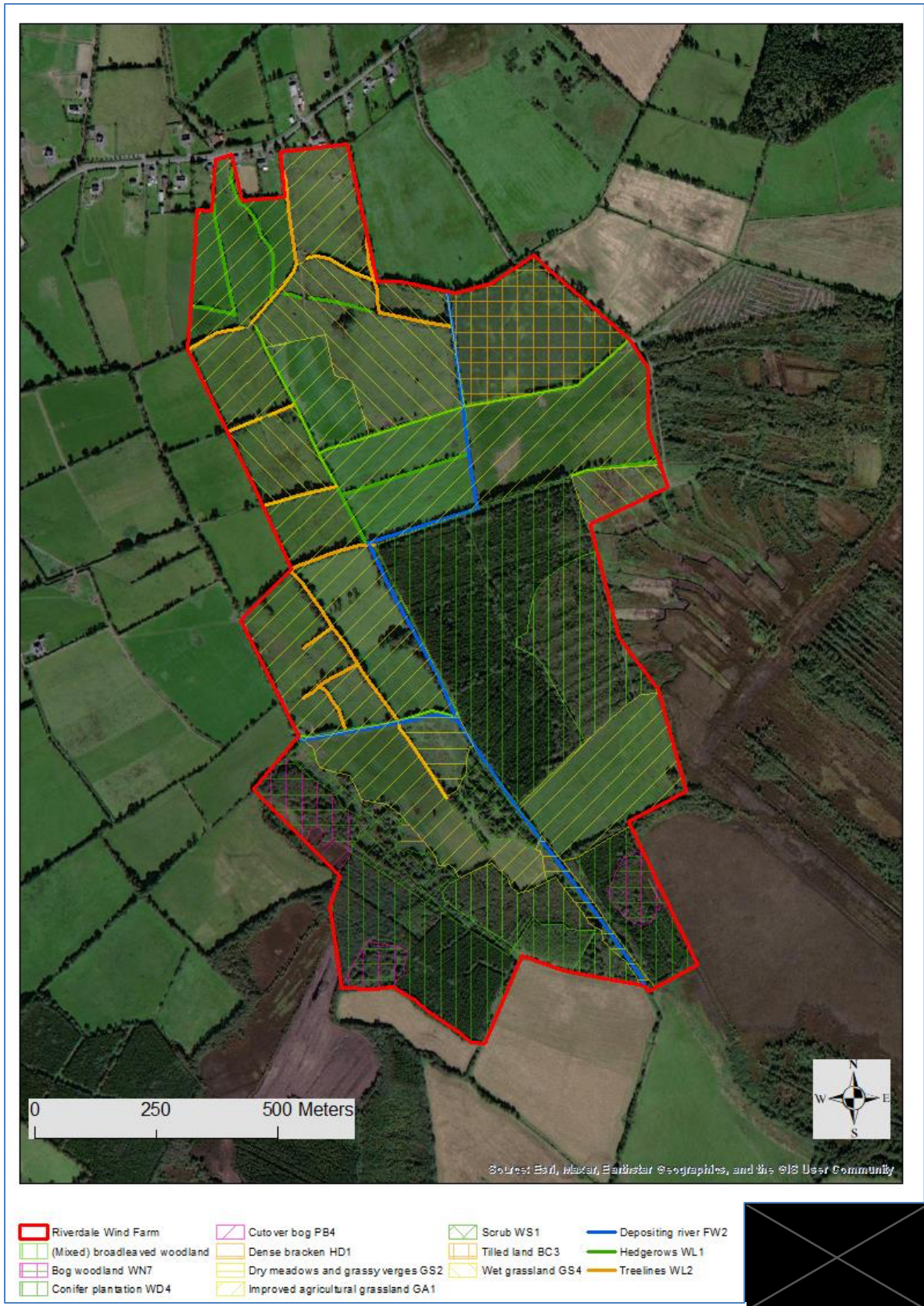


Figure 2: Preliminary habitat map of the potential wind farm site at

3.3 FLORA AND INVASIVE SPECIES

Based on the habitats present in the area it seems unlikely that there would be any presence of rare flora occurring within the site.

No high impact non-native species were recorded during the walkover survey within the potential wind farm site. Furthermore, no infestations of invasive species were recorded in proximity to the survey area.

3.4 FAUNA

Records of waterbirds at wetland sites in Ireland are monitored by the Irish Wetland Bird Survey (I-WeBS). A review of the mapped I-WeBS sites in Ireland was undertaken to determine their location in relation to the Riverdale site. Croboy Lough and Fields (OVS23) is the nearest I-WeBS site to the proposed, occurring circa 4km south. Whooper swans have been recorded at Croboy Lough as they forage in the area. According to the I-WeBS website, the mean number of whooper swans recorded at Croboy is 72 and the maximum number of whooper swans recorded was 129. These mean figures are based on the most recent recordings from period of 2015/17 – 2019/20. Other birds of interest include the Greylag goose which has a mean of 5 records, and a maximum of 13 in one season (figures from 2017/18 – 2018/19). A whooper swan desktop study undertaken in February 2021 (see Appendix I) presents information on whooper swan populations that occur in proximity to the Riverdale site. The outcome of the desktop review suggests that a whooper swan population roosts overnight on the ponds associated with the Kilsaran Quarry circa 1.5km southwest of the wind energy site and regularly use the adjacent grasslands for foraging during the day. Bird surveys are ongoing at Riverdale, and whooper swan were recorded flying through the site on two occasions in October 2021. The data from surveys carried out in the winter season 2021-2022 will help to determine the usage of the site by this species.

Whooper swans are listed on Annex I of the EU Birds Directive, which is the highest level of conservation status afforded to birds. They are large waterbirds susceptible to collision risk with turbines. Other bird species of note recorded within the site during the ecological walkover survey included mallard and swallows both of which are amber listed, kestrel and snipe, both of which are red listed (Gilbert, Stanbury and Lewis, 2021³).

The nearest Special Protection Area (SPA) to the proposed development is the River Boyne and River Blackwater SPA (NPWS Site Code: 004232). The SPA occurs circa 1.9km to the south of the potential wind farm site at its nearest point and is also hydrologically connected circa 1.9km downstream via the Craddanstown Stream. The SPA is designated for the conservation of kingfisher, another rare species listed on Annex I of the EU Birds Directive. The kingfisher would not be directly affected by wind turbines, however, their foraging habitats could be susceptible a decline in water quality as a result of turbine

³ Gilbert G, Stanbury A and Lewis L (2021), "Birds of Conservation Concern in Ireland 2020 –2026". Irish Birds 9: 523—544

construciton. The kingfisher is known to use the river Boyne to fish and nest (NPWS, 2022). Only one record of kingfisher was obtained from the River Boyne in 2010/11, no subsequent records have been obtained.

No observations of otter activity were noted at the Riverdale site. The Craddanstown Stream has potential for otter activity, but targeted surveys would be required to determine if there is any current presence of otters. It has however been documented that otters use the River Boyne for foraging and dwelling areas. Any disturbance to watercourses in the proposed wind farm site may have negative effects further downstream, potentially impacting otter prey availability.

The mature deciduous treelines within the site, reaching approximately 15 – 20m in height, may support bat species. The mature trees are almost all entirely covered in Ivy, which provides suitable habitat for bat and bird roosts. Additionally, there are two derelict farm sheds also located within this field. It is likely that bats use the trees in summer and the sheds in winter when needed. The bat suitability index suggests that the Riverdale site is suitable for eight resident bat species. There was no clear evidence of bats in this area at the time of the survey, however, targeted surveys are recommended to determine presence of bats. The sheds were also inspected for presence of barn owls. There were no signs of barn owl activity e.g., direct sightings, nests, pellets, etc.

3.4.1 Badger

Active badger setts were recorded within the potential wind farm site, see Figure 3. Two setts were found in the north of the site, and another to the north-west. Both setts occurred in similar settings (treelines/hedgerows at improved pasture boundaries) and seemed to be in use, as indicated by the presence of recently dug spoil heaps. A sett with two entrances was recorded in the central conifer plantation. The sett was located on a settled spoil heap excavated from a drainage ditch. The sett was considered active but not used for breeding. This was determined by the presence of distinct routes in/out of the sett, indicated by a disturbed surface and badger tracks into the tunnels and a scarcity in bedding material being gathered near or at the entrance of the sett. An active sett with two entrances was recorded at the south east of the site. The sett was determined to be an active breeding sett due to the freshly dug spoil heaps outside of the entrances with the addition of an abundance of bedding material at the sett entrance. Two Badger setts were also recorded within a dry linear ditch which extends diagonally south the west for circa 400m. The more northern of these setts had two sett entrances which appeared to be in use (regularly used entrances). The more southerly sett displayed a trail of bedding leading into the entrance. This suggests that this sett is actively being used for breeding.

Although no other badger setts were recorded in this area it is likely that they do occur in neighbouring areas to this site. Small tracks and latrines were recorded along hedgerows and treelines near the borders of the proposed site.

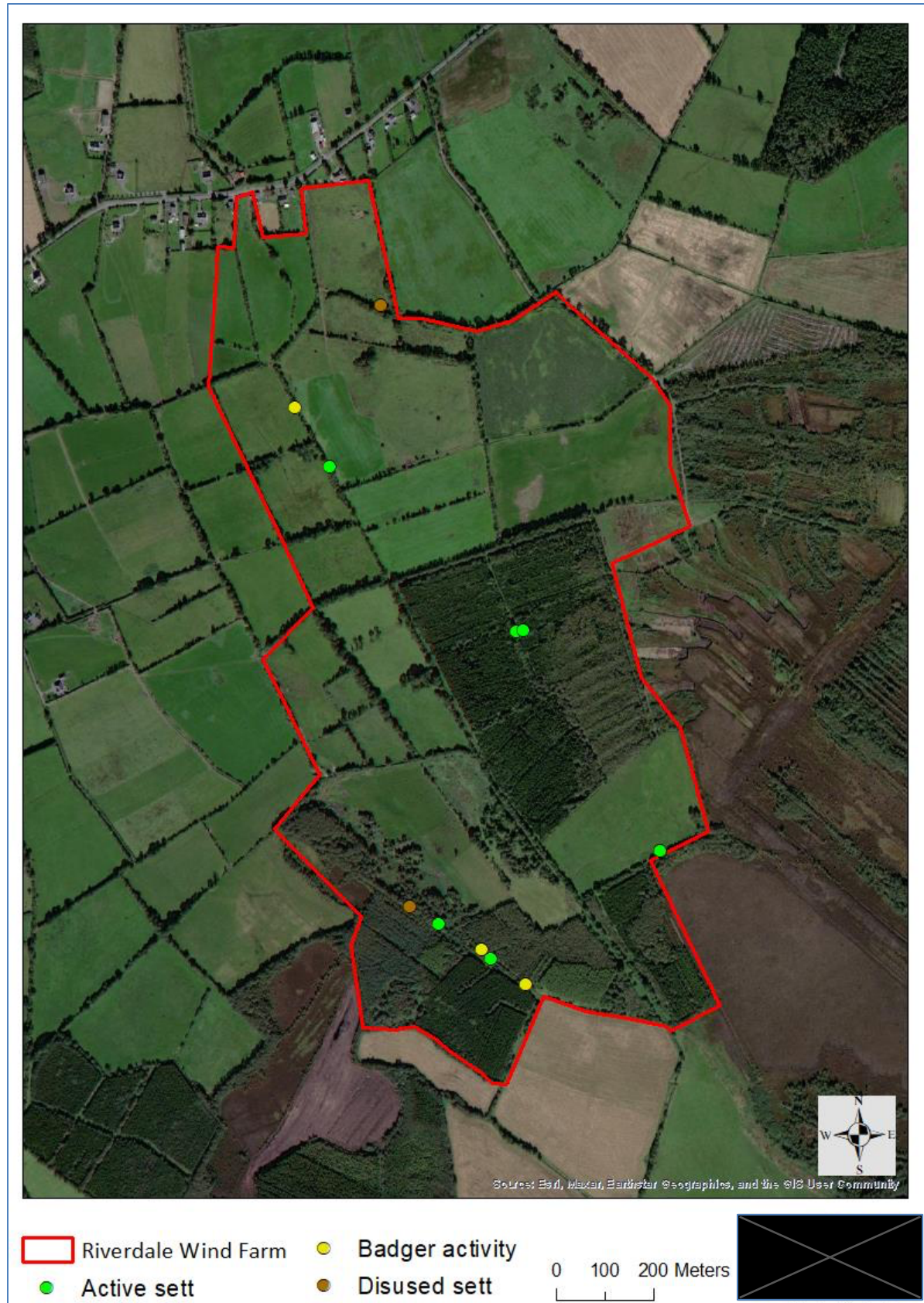


Figure 3: Badger activity recorded at the

4 IDENTIFICATION AND ASSESSMENT OF ECOLOGICAL CONSTRAINTS

Key ecological constraints identified during the preliminary appraisal are listed in Table 2.

Table 2: Key ecological constraints within the potential wind farm site.

Key ecological constraint	Evaluation	Recommendation	Mitigation / Recommendation
Craddanstown and Grange More Streams	International Importance	Avoid development within 50m of all natural watercourses.	Minimise risk of run-off during development through the implementation of silt and pollution control measures. Careful consideration for the management of excavated material.
Hedgerows and treelines	Local importance	Development should be setback from hedgerow / treeline field boundaries. Use existing gaps in hedgerows where possible.	Hedgerows are important linear features for birds, and other species such as badgers, which use them to traverse their foraging area.
Badger setts	Protected species Wildlife Act (1976) as amended	Avoid excavations / development within 50m of active setts.	Avoid development within proximity to badger setts. Provision of badger proof fencing and underpasses over roadways to foraging areas.
Birds	Unknown	Undertake 2 years annual surveys to determine occurrence of species of conservation concern.	To be determined (surveys are ongoing). Avoidance of development on semi-natural habitat of value to bird species.
Bats	National Importance, Protected species under the Wildlife Act (1976) as amended	Conduct targeted bat surveys within the suggested habitat area.	Avoid development near the area and do not remove or replace the existing habitat.

5 DISCUSSION AND CONCLUSIONS

The dominant habitats within the proposed wind farm site are of low ecological value. The watercourses that occur within the site occur upstream of European sites and so are of high ecological importance. In addition, the treelines and hedgerows within the site provide suitable habitat for bird and bat species, otter may be associated with watercourses within the site and badgers are active within the site. The key



ecological constraints and recommendations are outlined in Table 2 and the following surveys are recommended at the site:

- Continuation of bird surveys to compile 2 years of data (to follow SNH guidelines on assessment of avifauna at onshore wind farm sites).
- Bat surveys.
- Summer season botanical and habitat survey to include checks for invasive species.
- Aquatic ecology of watercourses at and downstream of site.



APPENDIX I

WHOOOPER SWAN REPORT

Document Control Sheet

Client	Highfield Energy
Project Title	Riverdale Wind Farm
Document Title	Whooper Swan Desktop Review
Project Number	WS0696
Document No.	RiverdaleWF_WS_Desktop

Revision	Status	Author	Review	Approved	Date
001	Draft	DOHD	BK	PC	05/02/2021
002	FINAL			PC	09/03/2021
					

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1 INTRODUCTION

[REDACTED] d. were commissioned by [REDACTED] assess the likely occurrence of Whooper Swan (*Cygnus cygnus*) in proximity to a potential wind energy site at [REDACTED]

[REDACTED] The potential interaction of a local population of Whooper Swans with the potential wind energy site is assessed based on existing data on occurrence and distribution.

1.1 Whooper Swans

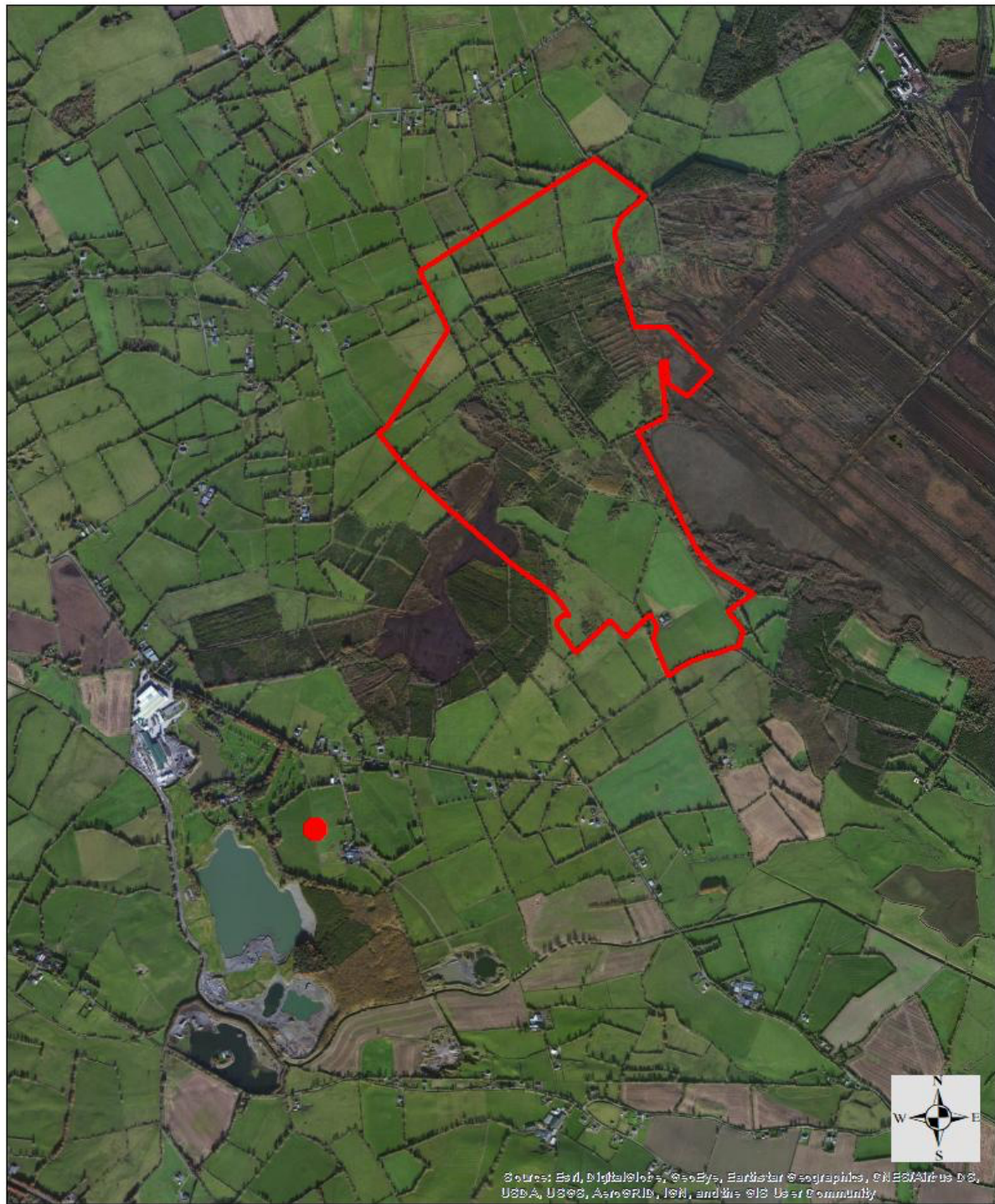
Whooper Swans (*Cygnus cygnus*) are a winter visitor to wetlands throughout Ireland from October to April. The Whooper Swans that are present in Ireland each winter breed in Iceland during the summer. Each year a small number of swans stay in Ireland for the summer and there have been occasional breeding records on lakes in the midlands and north-west. Whooper Swans predominantly winter on lowland open farmland around inland wetlands, and are regularly seen while feeding on grasslands and stubble. Whooper Swans are listed as Amber conservation concern (medium concern) under the Birds of Conservation Concern in Ireland (BoCCI) (2014-2019). As a result of their low flight line, Whooper Swans are considered to be at risk of collision with energy distribution infrastructure (overhead lines) or wind turbines. Displacement due to the presence of wind farm infrastructure is also seen as a potential risk where developments occur within suitable habitat.

1.2 Study Area

The potential wind energy site is a low-lying area located just west of Ballivor bog, Killucan, County Westmeath. Based on an interpretation of aerial photography the site itself includes areas of managed farmland (improved and semi-improved grassland), conifer plantation, and cutover bog. An extensive area of cutover raised bog (industrially cut) adjoins the site to the east. This bog appears to include a mosaic of uncut bog at the margin, regenerating cutover, and recently abandoned cutover. Intensively managed farmland dominates the landscape to the west, north, and south.

A local landowner reported a record of Whooper Swan foraging in a grassland area adjacent to a limestone quarry approximately 1km to the south-west of the potential wind farm site.

A map showing the location of the potential wind farm site and the nearby whooper swan feeding site is shown in Figure 1.



-  Riverdale Site Boundary
-  Whooper Swan Foraging

0 375 750 Meters

Figure 1: Study Area overlain on aerial imagery.

2 METHODOLOGY

This report was prepared following a review of existing ecological datasets relevant to the study area and following informal consultation with ornithological experts familiar with the area.

A desktop review of the site and surrounding habitats has been carried out by examining data from the National Biodiversity Data Centre, National Parks and Wildlife Service, Irish Wetland Bird Survey, and Westmeath County Council. See bibliography at rear of document for all data collected and examined for this desktop review.

3 RESULTS

3.1 Designated Sites

All sites designated for Nature Conservation within a 15km radius of the potential wind energy site are listed in Table 1 and illustrated in Figure 2.

The nearest Special Protection Area (SPA) to the proposed development, the River Boyne and River Blackwater SPA (NPWS Site Code: 004232), occurs circa 1km to the south of the potential wind energy site at its nearest point. The SPA is designated for the conservation of Kingfisher (*Alcedo atthis*), a species listed on Annex I of the EU Birds Directive. The Curris River which occurs within the study area flows to the SPA circa 1.1km downstream of the proposed development. The occurrence of designated sites in proximity to the proposed development is illustrated Table 1 below.

Table 1: Sites designated for nature conservation occurring within 15 of potential wind farm site.

NPWS Site code	NPWS Site name	Site description and location
004232	River Boyne and River Blackwater SPA	Long, linear site that comprises stretches of the River Boyne and several of its tributaries. The site is a Special Protection Area (SPA) under the E.U. Birds Directive of special conservation interest for Kingfisher (<i>Alcedo atthis</i>). Located 1km to the south of the potential wind energy site.
002205	Wooddown Bog SAC	Site is designated as a Degraded Raised Bog. Located circa 12.7km west of the potential wind energy site.
000694	Wooddown Bog NHA	Site is designated for Raised Bog. Located circa 12.7km west of the potential wind energy site.
002323	Milltownpass Bog NHA	Site is designated for Raised Bog. Located circa 12.5km to the south west of the potential wind energy site.

3.2 Other waterbird sites

A review of Irish Wetland Bird Survey (IWeBS) sites was undertaken. Three IWeBS sites occur within a 15km radius of the proposed wind energy site (see Table 2 below). Summary data for a nine year period from 2008 / 09 to 2017/18 was reviewed for the I-WeBS site Croboy Lough and Fields. The mean number of Whooper Swan counts during that period for this site is 55 swans with a maximum count of 129 individuals during the 2012 / 13 season. A total of 58 swans were recorded in the most recent data available for the site (2017 / 18). No observations of Whooper Swans were recorded in the remaining I-WeBS sites within a 15km radius of the Study Area

Table 2: Bird I-WeBS sites within 15km radius of the potential wind energy site.

I-WeBS Site Code	I-WeBS Site Name	Site Location	Whooper Swan Records
OVS23	Croboy Lough & fields	3km to the south of the proposed wind energy site.	Yes (mean number 2008-18 : 55indiv)
OW023	Crowinstown Lough	Circa 11.5km north of the proposed wind energy site.	No
OW024	Ballinlough	Circa 13.3km north of the proposed wind energy site.	No

Other wetland birds, waders, and species of conservation concern recorded in the three I-WeBS sites in proximity to the study area are summarized in

Table 3 below.

Table 3: Bird species recorded from I-WeBS sites in proximity to the Study Area.

Species	IWeBS Sites		
	Croboy Lough & fields	Crowinstown Lough	Ballinlough
<i>Grey Heron</i>		1	1
<i>Greylag Goose</i>	13		
<i>Kingfisher</i>			1
<i>Lapwing</i>		48	
<i>Little Grebe</i>	6		8
<i>Mallard</i>	12	75	710
<i>Moorhen</i>		2	15
<i>Mute Swan</i>	8		8
<i>Shoveler</i>		5	
<i>Teal</i>			91
<i>Wigeon</i>			90
<i>Whooper swan</i>	55		

3.3 Previous waterbird records

The proposed development occurs within the 10km grid square N65. No previous whooper swan records have been reported to the National Biodiversity Data Centre within this grid square (NBDC, 2021). A summary of bird species of conservation concern previously recorded within the 10km grid square is presented in Table 4 below.

Table 4: Bird species of conservation concern within the 10km grid square N65.

10km grid square	Species	Source	Conservation
N65	Curlew (<i>Numenius arquata</i>)	Bird Atlas 2007 -11	Annex I
	Lapwing (<i>Vanellus vanellus</i>)	Bird Atlas 2007 -11	Red listed
	Tufted Duck (<i>Aythya fuligula</i>)	Bird Atlas 2007 -11	Red listed

No information on the occurrence of Whooper Swans in the area was identified following a review of recent planning applications submitted to Westmeath County Council.

3.4 Consultation

The outcome of consultation with ornithological experts familiar with the area is summarised as follows:

- There is a population of Whooper Swans associate with the Kilsaran Concrete facility and they use the field as indicated in Figure 3below.
- The swans are not restricted to the field identified in Figure 3 but also a use other similar habitat in the area and their distribution can vary throughout a season and between years.
- At night the swans are known to roost on the quarry ponds.
- Up to 60+ swans is regular but numbers can exceed 100.
- The wider study area is generally of low value to bird species of conservation concern. Following wet weather, wet pastures fields can pull in Lapwing, gulls, and other water birds.
- Previous studies in the area did not identify birds traversing the bogs to the west of the potential wind energy site.
- Surveys would need to assess distribution a baseline study. Topography is hilly and undulating and the swans can be difficult to see from the roads therefore any future survey methods would need to account for this.

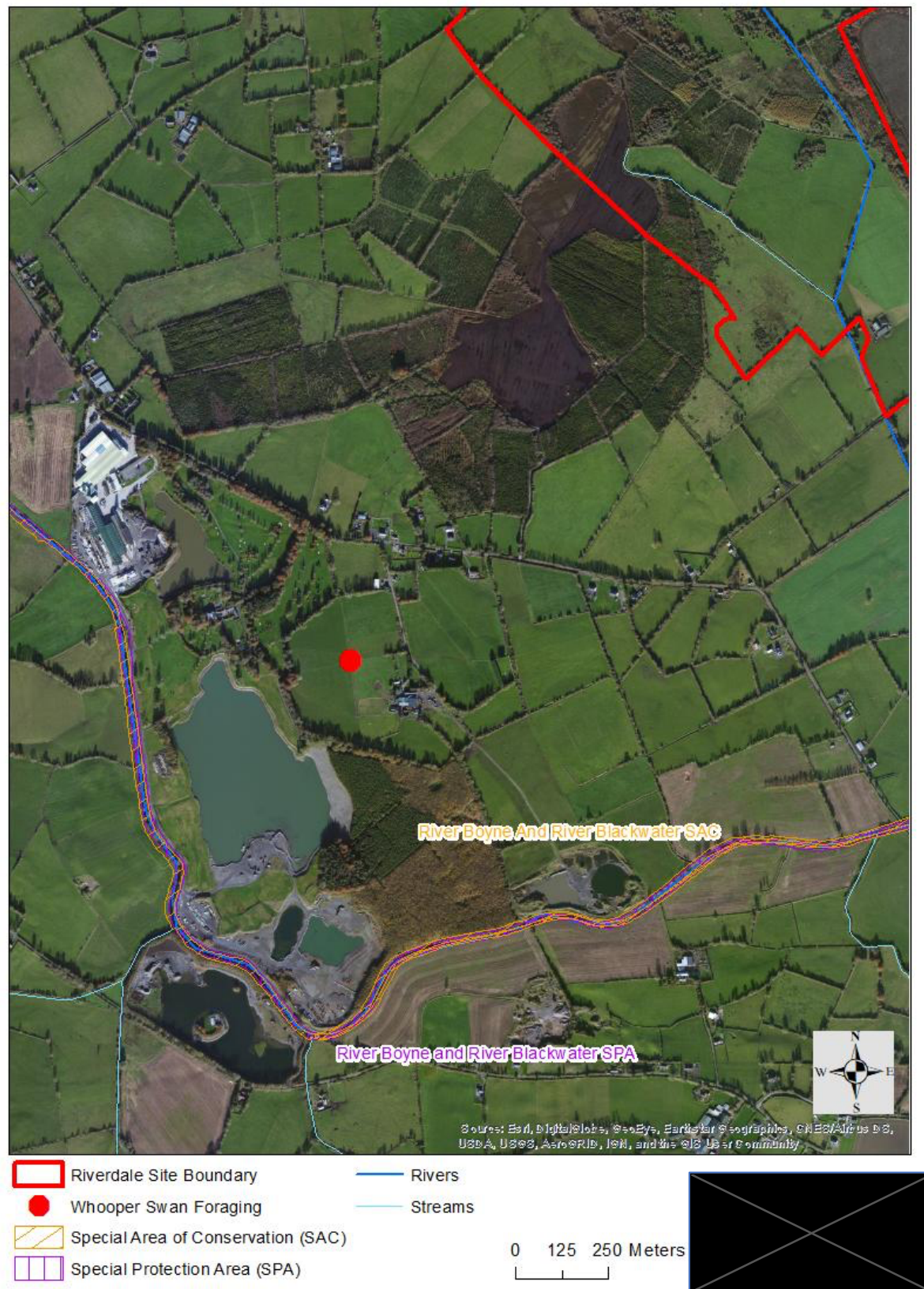


Figure 3: Whooper Swan foraging and roosting habitat in relation to European sites.

4 CONCLUSION

This desktop review of the study area confirmed the presence of a significant Whooper Swan population during winter. The population size is estimated at between 60 and 100 individuals and is therefore considered to be of national significance.

The outcome of consultation suggests that the swans roost overnight on the ponds associated with the Kilsaran Quarry ca 1.5km southwest of the wind energy site and regularly use the adjacent grasslands for foraging during the day. There is no data available to suggest whether the swans interact (use for feeding or fly over) with the potential wind energy site.

In the case of a wind energy facility being proposed in the area, it is recommended that extensive baseline surveys be undertaken in accordance with guidelines on undertaking baseline bird surveys at onshore wind farms (e.g. Scottish Natural Heritage 2017). Particular attention should be given to the distribution and movements of water birds in the wider study area throughout the winter and migration period.

5 BIBLIOGRAPHY

- Bird Atlas 2007-11 (Balmer, D., Gillings, S., Caffrey, B., Swann, B., Downie, I. & Fuller, R. Bird Atlas 2007-11: The breeding and wintering birds of Britain & Ireland. BTO Books, Thetford).
- Birdwatch Ireland (2015). Whooper swan distribution and breeding. Birdwatch Ireland.
- Crowe, O. (2005). Ireland's Wetlands and their waterbirds: Status and distribution. BirdWatch Ireland, Newcastle, County Wicklow.
- Irish Wetland Bird Survey (I-WeBS)(2021). Map of I_WeBS sites. Birdwatch Ireland.
- Lewis, L. J., Burke, B., Fitzgerald, N., Tierney, T. D. & Kelly, S. (2019) Irish Wetland Bird Survey: Waterbird Status and Distribution 2009/10-2015/16. Irish Wildlife Manuals, No. 106. National Parks and Wildlife Service, Department of Culture, Heritage and the Gaeltacht, Ireland.
- NBDC (2021). Biodiversity Maps. National Biodiversity Data Centre.
- NPWS (2002) Woododwn Bog NHA 000694: Site synopsis. National Parks and Wildlife Service, Department of Culture, Heritage and the Gaeltacht, Ireland.
- NPWS (2016) Wooddown Bog SAC 002205: Site synopsis. Revised version. National Parks and Wildlife Service, Department of Culture, Heritage and the Gaeltacht, Ireland.
- NPWS (2020) Conservation objectives for River Boyne and River Blackwater SPA [004232]. Generic Version 7.0. Department of Culture, Heritage and the Gaeltacht.
- Scottish Natural Heritage (2017). Recommended bird survey methods to inform impact assessment of onshore wind farms.
- Trinder, M. (2012). The potential consequences of elevated mortality on the population viability of whooper swans in relation to wind farm development in Northern Scotland. *Scottish Natural Heritage Commissioned Report No. 459*
- Westmeath CoCo (2021). Planning applications and urban development plans. Westmeath County Council.